Nuzhat Khan

AI / GenAI Engineer | Python, Vector Search (Qdrant), CLIP, LoRA, FastAPI

Mumbai, India (open to remote) | +91-9769149366 | knuzhat136@gmail.com
linkedin.com/in/nuzhat-khan-dev | github.com/Naazkn13 | nuzhat-portfolio-alpha.vercel.app

PROFESSIONAL SUMMARY

Python developer with 14 months of production backend experience in regulated BFSI software and hands-on applied AI. Built a multimodal semantic search engine (CLIP, Qdrant), implemented LoRA fine-tuning from scratch, and added hybrid re-ranking and LLM-assisted query expansion (Gemini). Strong on the engineering around models: document extraction pipelines, data validation, async APIs and containerised deployment. Available immediately.

TECHNICAL SKILLS

AI and GenAI: Embeddings, semantic and vector search, CLIP (open_clip ViT-B/32), hybrid re-ranking, LLM integration (Gemini API), query expansion, prompt engineering, RAG, LoRA fine-tuning, linear probe, adapter heads, Bayesian updating
Vector Databases: Qdrant (HNSW tuning), FAISS, Pinecone
Python and Data: Python, PyTorch, Pandas, NumPy, PyMuPDF, Tabula-py, Pydantic
Backend and APIs: FastAPI, REST APIs, async APIs, SQL, SQLAlchemy, PostgreSQL, Supabase, PyTest
DevOps: Docker, Docker Compose, Azure DevOps (CI/CD), GitHub Actions, Git, Linux
Frontend (working knowledge): React, Next.js
AI-Assisted Development and Agents: Claude, GPT, OpenCode, Hermes agent (Telegram and Discord integrations, Docker and server management, code review), agent memory and markdown agent-kit files, CodeRabbit

AI PROJECTS

FashionGallery: Multimodal Semantic Search Engine | Python, FastAPI, open_clip (ViT-B/32), Qdrant, Gemini, Docker | github.com/Naazkn13/FashionGallery

- Built joint image and text retrieval with CLIP embeddings indexed in Qdrant using tuned HNSW parameters (m=16, ef_construct=200), served through async FastAPI endpoints and packaged with Docker Compose.
- Implemented three fine-tuning approaches on top of CLIP: LoRA written from scratch (low-rank adapters with zero-initialised B matrix), a linear probe and an adapter head.
- Built a hybrid re-ranker that combines CLIP similarity with a custom dominant-colour extractor (HSV binning), and a Gemini-powered query expansion endpoint with a local fallback.
- Built the first version on FAISS and Pinecone, then rebuilt it on Qdrant.

Project Phoenix: Online Learning Layer for a Trading System | Python, Bayesian inference, SQLAlchemy, SQLite, pytest

- Built a learning layer that updates win-rate estimates after every closed trade (conjugate Beta Bayesian calibration), classifies failed trades (friction drag, false breakout, chop, timeout) and benches underperforming symbols for 24 hours.
- Persisted a trade journal with SQLAlchemy and covered the system with 37 pytest tests. Private repository; code walkthrough available on request.

PROFESSIONAL EXPERIENCE

Software Developer, Infomatics Services Pvt. Ltd., Mumbai | Jun 2025 – Jul 2026

- **CAS Statement Parser (client-deployed):** Built a Python document-extraction pipeline (PyMuPDF, Tabula-py, Pandas) packaged as a desktop app with Tauri and PyInstaller. Parsed unstructured CDSL statements at 98% accuracy on 500+ statements per month using rule-based layout extraction, saving 10+ hours per week.
- **Compulse (SEBI reporting platform):** Co-developed a platform automating 67+ SEBI report types across 43 maker-checker workflow stages with 32+ automated emails; built Pydantic validation schemas across heterogeneous data sources. Reached institutional client UAT.
- **Novus Comply and Novus UPSI:** Built a trade pre-clearance engine reconciling four data feeds (ECAS, BENPOS, KFin, CAMS) with SEBI PIT restricted-list checks, RBAC, Fernet-encrypted UPSI fields and tamper-evident audit logging, on async FastAPI endpoints over indexed PostgreSQL queries.
- **Employee Operations Portal:** Delivered a full-stack portal for 50+ employees with approval workflows and escalation rules, cutting HR processing overhead by 60%.
- Containerised services with Docker and shipped through Azure DevOps CI/CD pipelines to client UAT environments.

OTHER PROJECTS

Biometric Attendance and Payroll System | FastAPI, React, Supabase, Expo, ZKTeco SDK | attendance-sigma-one.vercel.app

- Real-time ZKTeco biometric sync and automated 30-day payroll with leave accrual, overtime and PDF payslips for an eye hospital (12+ staff); cut monthly processing from 2+ hours to under 30 minutes.

Hospital SOP and Operations Portal | Next.js, Python (ReportLab), Supabase

- Built for an eye hospital: a portal with role-based access, backed by a Python pipeline that generates 40+ SOP documents across 15 categories as PDFs.

EDUCATION

BSc Information Technology, Bhavan's College, Mumbai University | 2021 – 2024

CERTIFICATIONS

Advanced Program in Java Full Stack Development, DVOC Institute (Skill India / NSDC) | Dec 2025